Report on Assignment 2

Ivan Savelev DS-02

OVERALL ARCHITECTURE

The system follows a batch indexing and online query pattern suited to Hadoop and a distributed document collection:

Raw documents → HDFS → MapReduce (index + stats) → HDFS outputs
→ PySpark load → Cassandra/ScyllaDB → PySpark BM25 query → top-10 results

HDFS stores both the corpus (under /data and consolidated /input/data) and intermediate MapReduce outputs (/indexer/index, /indexer/doc_stats). Cassandra holds query-serving tables so searches do not scan HDFS. The stack is orchestrated in Docker Compose (Spark/YARN, HDFS, Cassandra) with shell entrypoints that run preparation, indexing, and demo queries end-to-end.

DATA PREPARATION

MapReduce streaming expects line-oriented text; the corpus may start as Parquet or many small files. prepare_data.py samples or materializes documents into a local data/ tree; consolidate_data.py reads whole-text files from HDFS /data and writes a single consolidated dataset under /input/data (tab-separated doc_id, title, body) so downstream jobs have one input pattern and predictable splits.

MAPREDUCE INDEXING (HADOOP STREAMING)

I use two separate pipelines instead of one monolithic job:

Pipeline 1 — Inverted index (mapper1 / reducer1)

- Mapper: tokenizes document text with the same rule as search (lowercase alphanumeric tokens via regex) and emits (term, doc_id).
- Reducer: aggregates per-term postings and term statistics needed for ranking (term frequency per document, document frequency), writing compact lines to HDFS for later loading into Cassandra.

Pipeline 2 — Document statistics (mapper2 / reducer2)

- Emits per-document metadata (e.g. identifier, title, document length) for BM25 length normalization and result display.

A single reduce task per pipeline keeps one merged part file per output, which simplifies loading small and medium corpora into Cassandra without merge logic in the loader. Hadoop Streaming runs Python mappers/reducers so the team can iterate quickly; the streaming JAR is resolved from the real tools/lib location (excluding *-sources* / *-test* jars) to avoid mis-invoking RunJar.

STORAGE LAYER

Design choice: normalize data into relational-style tables that match access patterns:

- vocabulary(term PRIMARY KEY, df) — quick lookup of df per query term.
- inverted_index(term, doc_id) — postings with tf; partition by term so all postings for a term are co-located for range reads.
- doc_stats(doc_id PRIMARY KEY, title, dl) — titles and lengths for BM25.
- corpus_stats(id PRIMARY KEY, num_docs, avg_dl) — global N and average document length for IDF and length normalization.

SimpleStrategy with replication factor 1 matches a single-node demo cluster. store_index.py reads MapReduce outputs from HDFS with Spark (local driver mode), parses lines, and issues batched inserts; connection retries tolerate slow Cassandra startup.

QUERY AND RANKING

Queries are tokenized consistently with indexing (same regex, lowercasing). For each query term, the client reads df from vocabulary and all (doc_id, tf) from inverted_index, then loads doc_stats only for candidate documents.

BM25 is implemented with standard parameters (k1 = 1.5, b = 0.75): IDF uses the corpus document count N and df; term frequency is normalized by document length dl and corpus average length avg_dl. Scores for multiple terms are summed per document in Spark (parallelize → map → reduceByKey). The top 10 documents by score are returned with titles from doc_stats.

This split—Cassandra for structured retrieval of postings and stats, Spark for score aggregation—keeps the hot path flexible and uses Spark where parallel aggregation helps when many postings exist.

Note: in the work the problem related to Cassandra connection occurred. It is hard for me to tell exactly why it occurs. Still, you can see query processing implementation that should otherwise work.

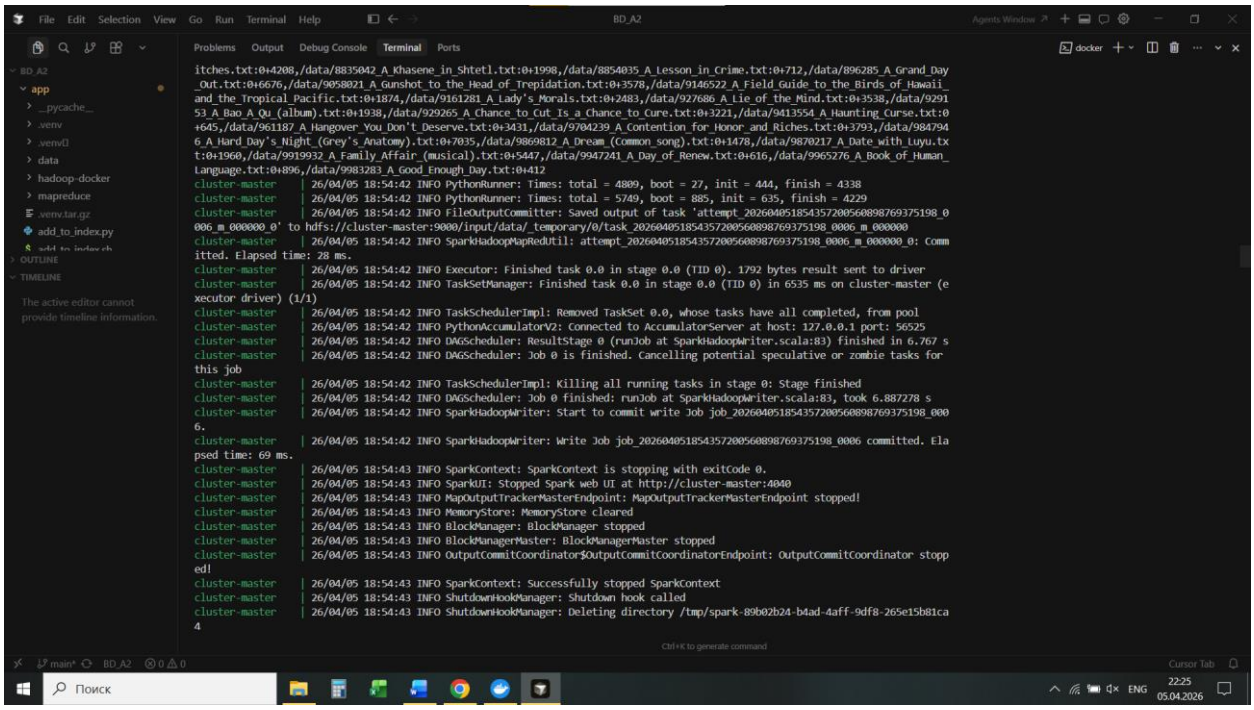
OPERATIONAL CHOICES

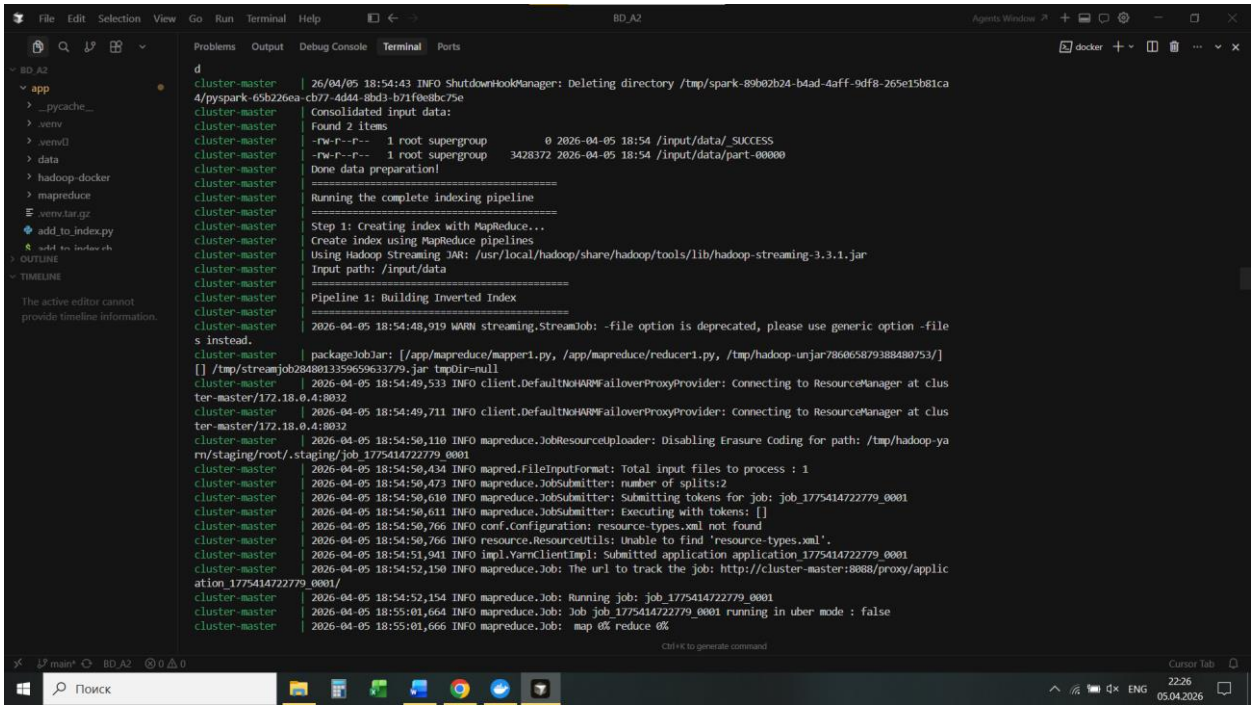
- YARN spark-submit with a packed virtual environment ensures executor Python matches the driver for query jobs.
- Waits and retries on Cassandra (socket readiness and driver connect) reduce races during container startup under memory pressure; JVM heap limits for Cassandra mitigate OOM (exit 137) on constrained hosts.

DEMONSTRATION

In order to execute code run “docker compose up” from root of repository.

Here you can see second version of execution. In order to check first, see “screenshots” folder





```
cluster-master | 2026-04-05 18:55:08,790 INFO mapreduce.Job:  map 100% reduce 0%
cluster-master | 2026-04-05 18:55:13,819 INFO mapreduce.Job:  map 100% reduce 100%
cluster-master | 2026-04-05 18:55:14,831 INFO mapreduce.Job: Job job_1775414722779_0001 completed successfully
cluster-master | 2026-04-05 18:55:14,916 INFO mapreduce.Job: Counters: 55
cluster-master |     File System Counters
cluster-master |             FILE: Number of bytes read=9127582
cluster-master |             FILE: Number of bytes written=19084618
cluster-master |             FILE: Number of read operations=0
cluster-master |             FILE: Number of large read operations=0
cluster-master |             FILE: Number of write operations=0
cluster-master |             HDFS: Number of bytes read=3432668
cluster-master |             HDFS: Number of bytes written=2981600
cluster-master |             HDFS: Number of read operations=11
cluster-master |             HDFS: Number of large read operations=0
cluster-master |             HDFS: Number of write operations=2
cluster-master |             HDFS: Number of bytes read erasure-coded=0
cluster-master |     Job Counters
cluster-master |             Killed map tasks=1
cluster-master |             Launched map tasks=2
cluster-master |             Launched reduce tasks=1
cluster-master |             Data-local map tasks=2
cluster-master |             Total time spent by all maps in occupied slots (ms)=9175
cluster-master |             Total time spent by all reduces in occupied slots (ms)=3098
cluster-master |             Total time spent by all map tasks (ms)=9175
cluster-master |             Total time spent by all reduce tasks (ms)=3098
cluster-master |             Total vcore-milliseconds taken by all map tasks=9175
cluster-master |             Total vcore-milliseconds taken by all reduce tasks=3098
cluster-master |             Total megabyte-milliseconds taken by all map tasks=9395200
cluster-master |             Total megabyte-milliseconds taken by all reduce tasks=3172352
cluster-master |     Map-Reduce Framework
cluster-master |             Map input records=978
cluster-master |             Map output records=562627
cluster-master |             Map output bytes=8002322
cluster-master |             Map output materialized bytes=9127588
cluster-master |             Input split bytes=200
cluster-master |             Combine input records=0
cluster-master |             Combine output records=0
cluster-master |             Reduce input groups=39127
cluster-master |             Reduce shuffle bytes=9127588
cluster-master |             Reduce input records=562627
```

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cluster-master |             IO_ERROR=0
cluster-master |             WRONG_LENGTH=0
cluster-master |             WRONG_MAP=0
cluster-master |             WRONG_REDUCE=0
cluster-master |     File Input Format Counters
cluster-master |             Bytes Read=3432468
cluster-master |     File Output Format Counters
cluster-master |             Bytes Written=2981600
cluster-master | 2026-04-05 18:55:14,916 INFO streaming.StreamJob: Output directory: /indexer/index
cluster-master | Pipeline 1 complete. Output:
cluster-master | Found 2 items
cluster-master | -rw-r--r--   1 root supergroup          0 2026-04-05 18:55 /indexer/index/_SUCCESS
cluster-master | -rw-r--r--   1 root supergroup    2981600 2026-04-05 18:55 /indexer/index/part-00000
cluster-master | ==========================================
cluster-master | Pipeline 2: Computing Document Statistics
cluster-master | ==========================================
cluster-master | 2026-04-05 18:55:18,189 WARN streaming.StreamJob: -file option is deprecated, please use generic option -file
s instead.
cluster-master | packageJobJar: [/app/mapreduce/mapper2.py, /app/mapreduce/reducer2.py, /tmp/hadoop-unjar2375311282744553633/]
 [] /tmp/streamjob2473902746728099093.jar tmpDir=null
cluster-master | 2026-04-05 18:55:18,799 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at clus
ter-master/172.18.0.4:8032
cluster-master | 2026-04-05 18:55:18,967 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at clus
ter-master/172.18.0.4:8032
cluster-master | 2026-04-05 18:55:19,126 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-ya
rn/staging/root/.staging/job_1775414722779_0002
cluster-master | 2026-04-05 18:55:19,386 INFO mapred.FileInputFormat: Total input files to process : 1
cluster-master | 2026-04-05 18:55:19,445 INFO mapreduce.JobSubmitter: number of splits:2
cluster-master | 2026-04-05 18:55:19,543 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1775414722779_0002
cluster-master | 2026-04-05 18:55:19,544 INFO mapreduce.JobSubmitter: Executing with tokens: []
cluster-master | 2026-04-05 18:55:19,687 INFO conf.Configuration: resource-types.xml not found
cluster-master | 2026-04-05 18:55:19,687 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
cluster-master | 2026-04-05 18:55:19,753 INFO impl.YarnClientImpl: Submitted application application_1775414722779_0002
cluster-master | 2026-04-05 18:55:19,784 INFO mapreduce.Job: The url to track the job: http://cluster-master:8088/proxy/applic
ation_1775414722779_0002/
cluster-master | 2026-04-05 18:55:19,786 INFO mapreduce.Job: Running job: job_1775414722779_0002
cluster-master | 2026-04-05 18:55:24,866 INFO mapreduce.Job: Job job_1775414722779_0002 running in uber mode : false
cluster-master | 2026-04-05 18:55:24,867 INFO mapreduce.Job:  map 0% reduce 0%
cluster-master | 2026-04-05 18:55:29,917 INFO mapreduce.Job:  map 100% reduce 0%
cluster-master | 2026-04-05 18:55:33,984 INFO mapreduce.Job:  map 100% reduce 100%
```

```
cluster-master | 2026-04-05 18:55:19,753 INFO impl.YarnClientImpl: Submitted application application_1775414722779_0002
cluster-master | 2026-04-05 18:55:19,784 INFO mapreduce.Job: The url to track the job: http://cluster-master:8088/proxy/applic
ation_1775414722779_0002/
cluster-master | 2026-04-05 18:55:19,786 INFO mapreduce.Job: Running job: job_1775414722779_0002
cluster-master | 2026-04-05 18:55:24,866 INFO mapreduce.Job: Job job_1775414722779_0002 running in uber mode : false
cluster-master | 2026-04-05 18:55:24,867 INFO mapreduce.Job:  map 0% reduce 0%
cluster-master | 2026-04-05 18:55:29,917 INFO mapreduce.Job:  map 100% reduce 0%
cluster-master | 2026-04-05 18:55:33,984 INFO mapreduce.Job:  map 100% reduce 100%
cluster-master | 2026-04-05 18:55:33,998 INFO mapreduce.Job: Job job_1775414722779_0002 completed successfully
cluster-master | 2026-04-05 18:55:34,100 INFO mapreduce.Job: Counters: 54
cluster-master |     File System Counters
cluster-master |             FILE: Number of bytes read=37038
cluster-master |             FILE: Number of bytes written=903542
cluster-master |             FILE: Number of read operations=0
cluster-master |             FILE: Number of large read operations=0
cluster-master |             FILE: Number of write operations=0
cluster-master |             HDFS: Number of bytes read=3432668
cluster-master |             HDFS: Number of bytes written=35076
cluster-master |             HDFS: Number of read operations=11
cluster-master |             HDFS: Number of large read operations=0
cluster-master |             HDFS: Number of write operations=2
cluster-master |             HDFS: Number of bytes read erasure-coded=0
cluster-master |     Job Counters
cluster-master |             Launched map tasks=2
cluster-master |             Launched reduce tasks=1
cluster-master |             Data-local map tasks=2
cluster-master |             Total time spent by all maps in occupied slots (ms)=4898
cluster-master |             Total time spent by all reduces in occupied slots (ms)=1535
cluster-master |             Total time spent by all map tasks (ms)=4898
cluster-master |             Total time spent by all reduce tasks (ms)=1535
cluster-master |             Total vcore-milliseconds taken by all map tasks=4898
cluster-master |             Total vcore-milliseconds taken by all reduce tasks=1535
cluster-master |             Total megabyte-milliseconds taken by all map tasks=5015552
cluster-master |             Total megabyte-milliseconds taken by all reduce tasks=1571840
cluster-master |     Map-Reduce Framework
cluster-master |             Map input records=978
cluster-master |             Map output records=978
cluster-master |             Map output bytes=35076
cluster-master |             Map output materialized bytes=37044
cluster-master |             Input split bytes=200
```

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cluster-master |     Map-Reduce Framework
cluster-master |             Map input records=978
cluster-master |             Map output records=978
cluster-master |             Map output bytes=35076
cluster-master |             Map output materialized bytes=37044
cluster-master |             Input split bytes=200
cluster-master |             Combine input records=0
cluster-master |             Combine output records=0
cluster-master |             Reduce input groups=978
cluster-master |             Reduce shuffle bytes=37044
cluster-master |             Reduce input records=978
cluster-master |             Reduce output records=978
cluster-master |             Spilled Records=1956
cluster-master |             Shuffled Maps =2
cluster-master |             Failed Shuffles=0
cluster-master |             Merged Map outputs=2
cluster-master |             GC time elapsed (ms)=131
cluster-master |             CPU time spent (ms)=2920
cluster-master |             Physical memory (bytes) snapshot=772755456
cluster-master |             Virtual memory (bytes) snapshot=7708610560
cluster-master |             Total committed heap usage (bytes)=642777088
cluster-master |             Peak Map Physical memory (bytes)=291954688
cluster-master |             Peak Map Virtual memory (bytes)=2572226560
cluster-master |             Peak Reduce Physical memory (bytes)=190779392
cluster-master |             Peak Reduce Virtual memory (bytes)=2570207232
cluster-master |     Shuffle Errors
cluster-master |             BAD_ID=0
cluster-master |             CONNECTION=0
cluster-master |             IO_ERROR=0
cluster-master |             WRONG_LENGTH=0
cluster-master |             WRONG_MAP=0
cluster-master |             WRONG_REDUCE=0
cluster-master |     File Input Format Counters
cluster-master |             Bytes Read=3432468
cluster-master |     File Output Format Counters
cluster-master |             Bytes Written=35076
cluster-master | 2026-04-05 18:55:34,100 INFO streaming.StreamJob: Output directory: /indexer/doc_stats
cluster-master | Pipeline 2 complete. Output:
cluster-master | Found 2 items
cluster-master | -rw-r--r--   1 root supergroup          0 2026-04-05 18:55 /indexer/doc_stats/_SUCCESS
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cluster-master    |             IO_ERROR=0
cluster-master    |             WRONG_LENGTH=0
cluster-master    |             WRONG_MAP=0
cluster-master    |             WRONG_REDUCE=0
cluster-master    |     File Input Format Counters
cluster-master    |             Bytes Read=3432468
cluster-master    |     File Output Format Counters
cluster-master    |             Bytes Written=35076
cluster-master    | 2026-04-05 18:55:34,100 INFO streaming.StreamJob: Output directory: /indexer/doc_stats
cluster-master    | Pipeline 2 complete. Output:
cluster-master    | Found 2 items
cluster-master    | -rw-r--r--   1 root supergroup          0 2026-04-05 18:55 /indexer/doc_stats/_SUCCESS
cluster-master    | -rw-r--r--   1 root supergroup      35076 2026-04-05 18:55 /indexer/doc_stats/part-00000
cluster-master    | ==========================================
cluster-master    | All MapReduce pipelines completed!
cluster-master    | ==========================================
cluster-master    | Found 2 items
cluster-master    | drwxr-xr-x   - root supergroup          0 2026-04-05 18:55 /indexer/doc_stats
cluster-master    | drwxr-xr-x   - root supergroup          0 2026-04-05 18:55 /indexer/index
cluster-master    | Step 2: Storing index in Cassandra...
cluster-master    | Store the index and others to Cassandra/ScyllaDB tables
cluster-master    | 26/04/05 18:55:40 INFO SparkContext: Running Spark version 3.5.4
cluster-master    | 26/04/05 18:55:40 INFO SparkContext: OS info Linux, 6.6.87.2-microsoft-standard-WSL2, amd64
cluster-master    | 26/04/05 18:55:40 INFO SparkContext: Java version 1.8.0_442
cluster-master    | 26/04/05 18:55:40 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using buil
tin-java classes where applicable
cluster-master    | 26/04/05 18:55:40 INFO ResourceUtils: ==============================================================
cluster-master    | 26/04/05 18:55:40 INFO ResourceUtils: No custom resources configured for spark.driver.
cluster-master    | 26/04/05 18:55:40 INFO ResourceUtils: ==============================================================
cluster-master    | 26/04/05 18:55:40 INFO SparkContext: Submitted application: Store Index
cluster-master    | 26/04/05 18:55:40 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> nam
e: cores, amount: 1, script: , vendor: , memory -> name: memory, amount: 1024, script: , vendor: , offHeap -> name: offHeap, amou
nt: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount: 1.0)
cluster-master    | 26/04/05 18:55:40 INFO ResourceProfile: Limiting resource is cpu
cluster-master    | 26/04/05 18:55:40 INFO ResourceProfileManager: Added ResourceProfile id: 0
cluster-master    | 26/04/05 18:55:40 INFO SecurityManager: Changing view acls to: root
cluster-master    | 26/04/05 18:55:40 INFO SecurityManager: Changing modify acls to: root
cluster-master    | 26/04/05 18:55:40 INFO SecurityManager: Changing view acls groups to:
cluster-master    | 26/04/05 18:55:40 INFO SecurityManager: Changing modify acls groups to:
cluster-master    | 26/04/05 18:55:40 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users wit
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cluster-master    | 26/04/05 18:55:40 INFO SparkEnv: Registering MapOutputTracker
cluster-master    | 26/04/05 18:55:40 INFO SparkEnv: Registering BlockManagerMaster
cluster-master    | 26/04/05 18:55:40 INFO BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for g
etting topology information
cluster-master    | 26/04/05 18:55:40 INFO BlockManagerMasterEndpoint: BlockManagerMasterEndpoint up
cluster-master    | 26/04/05 18:55:40 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
cluster-master    | 26/04/05 18:55:40 INFO DiskBlockManager: Created local directory at /tmp/blockmgr-fe859b39-13ff-4cb9-85ff-5e9
aaaa7740d
cluster-master    | 26/04/05 18:55:40 INFO MemoryStore: MemoryStore started with capacity 366.3 MiB
cluster-master    | 26/04/05 18:55:40 INFO SparkEnv: Registering OutputCommitCoordinator
cluster-master    | 26/04/05 18:55:41 INFO JettyUtils: Start Jetty 0.0.0.0:4040 for SparkUI
cluster-master    | 26/04/05 18:55:41 INFO Utils: Successfully started service 'SparkUI' on port 4040.
cluster-master    | 26/04/05 18:55:41 INFO Executor: Starting executor ID driver on host cluster-master
cluster-master    | 26/04/05 18:55:41 INFO Executor: OS info Linux, 6.6.87.2-microsoft-standard-WSL2, amd64
cluster-master    | 26/04/05 18:55:41 INFO Executor: Java version 1.8.0_442
cluster-master    | 26/04/05 18:55:41 INFO Executor: Starting executor with user classpath (userClassPathFirst = false): ''
cluster-master    | 26/04/05 18:55:41 INFO Executor: Created or updated repl class loader org.apache.spark.util.MutableURLClassLo
ader@21ed40 for default.
cluster-master    | 26/04/05 18:55:41 INFO Utils: Successfully started service 'org.apache.spark.network.netty.NettyBlockTransfer
Service' on port 39671.
cluster-master    | 26/04/05 18:55:41 INFO NettyBlockTransferService: Server created on cluster-master:39671
cluster-master    | 26/04/05 18:55:41 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block re
plication policy
cluster-master    | 26/04/05 18:55:41 INFO BlockManagerMaster: Registering BlockManager BlockManagerId(driver, cluster-master, 39
671, None)
cluster-master    | 26/04/05 18:55:41 INFO BlockManagerMasterEndpoint: Registering block manager cluster-master:39671 with 366.3
MiB RAM, BlockManagerId(driver, cluster-master, 39671, None)
cluster-master    | 26/04/05 18:55:41 INFO BlockManagerMaster: Registered BlockManager BlockManagerId(driver, cluster-master, 396
71, None)
cluster-master    | 26/04/05 18:55:41 INFO BlockManager: Initialized BlockManager: BlockManagerId(driver, cluster-master, 39671,
None)
cluster-master    | Reading inverted index from HDFS...
cluster-master    | 26/04/05 18:55:42 INFO MemoryStore: Block broadcast_0 stored as values in memory (estimated size 359.8 KiB, f
ree 365.9 MiB)
cluster-master    | 26/04/05 18:55:42 INFO MemoryStore: Block broadcast_0_piece0 stored as bytes in memory (estimated size 33.2 K
iB, free 365.9 MiB)
cluster-master    | 26/04/05 18:55:42 INFO BlockManagerInfo: Added broadcast_0_piece0 in memory on cluster-master:39671 (size: 33
.2 KiB, free: 366.3 MiB)
cluster-master    | 26/04/05 18:55:42 INFO SparkContext: Created broadcast 0 from textFile at NativeMethodAccessorImpl.java:0
cluster-master    | 26/04/05 18:55:42 INFO FileInputFormat: Total input files to process : 1
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e 365.9 MiB)
cluster-master    | 26/04/05 18:55:43 INFO MemoryStore: Block broadcast_1_piece0 stored as bytes in memory (estimated size 2.8 Ki
B, free 365.9 MiB)
cluster-master    | 26/04/05 18:55:43 INFO BlockManagerInfo: Added broadcast_1_piece0 in memory on cluster-master:39671 (size: 2.
8 KiB, free: 366.3 MiB)
cluster-master    | 26/04/05 18:55:43 INFO SparkContext: Created broadcast 1 from broadcast at DAGScheduler.scala:1585
cluster-master    | 26/04/05 18:55:43 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 0 (/indexer/index/part-* Map
PartitionsRDD[1] at textFile at NativeMethodAccessorImpl.java:0) (first 15 tasks are for partitions Vector(0))
cluster-master    | 26/04/05 18:55:43 INFO TaskSchedulerImpl: Adding task set 0.0 with 1 tasks resource profile 0
cluster-master    | 26/04/05 18:55:43 INFO TaskSetManager: Starting task 0.0 in stage 0.0 (TID 0) (cluster-master, executor drive
r, partition 0, ANY, 9066 bytes)
cluster-master    | 26/04/05 18:55:43 INFO Executor: Running task 0.0 in stage 0.0 (TID 0)
cluster-master    | 26/04/05 18:55:43 INFO HadoopRDD: Input split: hdfs://cluster-master:9000/indexer/index/part-00000:0+2981600
cluster-master    | 26/04/05 18:55:43 INFO MemoryStore: Block taskresult_0 stored as bytes in memory (estimated size 2.9 MiB, fre
e 363.0 MiB)
cluster-master    | 26/04/05 18:55:43 INFO BlockManagerInfo: Added taskresult_0 in memory on cluster-master:39671 (size: 2.9 MiB,
 free: 363.3 MiB)
cluster-master    | 26/04/05 18:55:43 INFO Executor: Finished task 0.0 in stage 0.0 (TID 0). 3075823 bytes result sent via BlockM
anager)
cluster-master    | 26/04/05 18:55:43 INFO TransportClientFactory: Successfully created connection to cluster-master/172.18.0.4:3
9671 after 25 ms (0 ms spent in bootstraps)
cluster-master    | 26/04/05 18:55:43 INFO TaskSetManager: Finished task 0.0 in stage 0.0 (TID 0) in 545 ms on cluster-master (ex
ecutor driver) (1/1)
cluster-master    | 26/04/05 18:55:43 INFO TaskSchedulerImpl: Removed TaskSet 0.0, whose tasks have all completed, from pool
cluster-master    | 26/04/05 18:55:43 INFO BlockManagerInfo: Removed taskresult_0 on cluster-master:39671 in memory (size: 2.9 Mi
B, free: 366.3 MiB)
cluster-master    | 26/04/05 18:55:43 INFO DAGScheduler: ResultStage 0 (collect at /app/store_index.py:33) finished in 0.671 s
cluster-master    | 26/04/05 18:55:43 INFO DAGScheduler: Job 0 is finished. Cancelling potential speculative or zombie tasks for
this job
cluster-master    | 26/04/05 18:55:43 INFO TaskSchedulerImpl: Killing all running tasks in stage 0: Stage finished
cluster-master    | 26/04/05 18:55:43 INFO DAGScheduler: Job 0 finished: collect at /app/store_index.py:33, took 0.718585 s
cluster-master    | Reading document stats from HDFS...
cluster-master    | 26/04/05 18:55:43 INFO MemoryStore: Block broadcast_2 stored as values in memory (estimated size 359.9 KiB, f
ree 365.6 MiB)
cluster-master    | 26/04/05 18:55:43 INFO MemoryStore: Block broadcast_2_piece0 stored as bytes in memory (estimated size 33.2 K
iB, free 365.5 MiB)
cluster-master    | 26/04/05 18:55:43 INFO BlockManagerInfo: Added broadcast_2_piece0 in memory on cluster-master:39671 (size: 33
.2 KiB, free: 366.2 MiB)
cluster-master    | 26/04/05 18:55:43 INFO SparkContext: Created broadcast 2 from textFile at NativeMethodAccessorImpl.java:0
cluster-master    | 26/04/05 18:55:43 INFO FileInputFormat: Total input files to process : 1
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cluster-master    | 26/04/05 18:55:43 INFO MemoryStore: Block broadcast_2_piece0 stored as bytes in memory (estimated size 33.2 K
iB, free 365.5 MiB)
cluster-master    | 26/04/05 18:55:43 INFO BlockManagerInfo: Added broadcast_2_piece0 in memory on cluster-master:39671 (size: 33
.2 KiB, free: 366.2 MiB)
cluster-master    | 26/04/05 18:55:43 INFO SparkContext: Created broadcast 2 from textFile at NativeMethodAccessorImpl.java:0
cluster-master    | 26/04/05 18:55:43 INFO FileInputFormat: Total input files to process : 1
cluster-master    | 26/04/05 18:55:43 INFO SparkContext: Starting job: collect at /app/store_index.py:36
cluster-master    | 26/04/05 18:55:43 INFO DAGScheduler: Got job 1 (collect at /app/store_index.py:36) with 1 output partitions
cluster-master    | 26/04/05 18:55:43 INFO DAGScheduler: Final stage: ResultStage 1 (collect at /app/store_index.py:36)
cluster-master    | 26/04/05 18:55:43 INFO DAGScheduler: Parents of final stage: List()
cluster-master    | 26/04/05 18:55:43 INFO DAGScheduler: Missing parents: List()
cluster-master    | 26/04/05 18:55:43 INFO DAGScheduler: Submitting ResultStage 1 (/indexer/doc_stats/part-* MapPartitionsRDD[3]
at textFile at NativeMethodAccessorImpl.java:0), which has no missing parents
cluster-master    | 26/04/05 18:55:43 INFO MemoryStore: Block broadcast_3 stored as values in memory (estimated size 4.9 KiB, fre
e 365.5 MiB)
cluster-master    | 26/04/05 18:55:43 INFO MemoryStore: Block broadcast_3_piece0 stored as bytes in memory (estimated size 2.8 Ki
B, free 365.5 MiB)
cluster-master    | 26/04/05 18:55:43 INFO BlockManagerInfo: Added broadcast_3_piece0 in memory on cluster-master:39671 (size: 2.
8 KiB, free: 366.2 MiB)
cluster-master    | 26/04/05 18:55:43 INFO SparkContext: Created broadcast 3 from broadcast at DAGScheduler.scala:1585
cluster-master    | 26/04/05 18:55:43 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 1 (/indexer/doc_stats/part-*
 MapPartitionsRDD[3] at textFile at NativeMethodAccessorImpl.java:0) (first 15 tasks are for partitions Vector(0))
cluster-master    | 26/04/05 18:55:43 INFO TaskSchedulerImpl: Adding task set 1.0 with 1 tasks resource profile 0
cluster-master    | 26/04/05 18:55:43 INFO TaskSetManager: Starting task 0.0 in stage 1.0 (TID 1) (cluster-master, executor drive
r, partition 0, ANY, 9070 bytes)
cluster-master    | 26/04/05 18:55:43 INFO Executor: Running task 0.0 in stage 1.0 (TID 1)
cluster-master    | 26/04/05 18:55:43 INFO HadoopRDD: Input split: hdfs://cluster-master:9000/indexer/doc_stats/part-00000:0+3507
6
cluster-master    | 26/04/05 18:55:43 INFO Executor: Finished task 0.0 in stage 1.0 (TID 1). 38190 bytes result sent to driver
cluster-master    | 26/04/05 18:55:43 INFO TaskSetManager: Finished task 0.0 in stage 1.0 (TID 1) in 30 ms on cluster-master (exe
cutor driver) (1/1)
cluster-master    | 26/04/05 18:55:43 INFO TaskSchedulerImpl: Removed TaskSet 1.0, whose tasks have all completed, from pool
cluster-master    | 26/04/05 18:55:43 INFO DAGScheduler: ResultStage 1 (collect at /app/store_index.py:36) finished in 0.043 s
cluster-master    | 26/04/05 18:55:43 INFO DAGScheduler: Job 1 is finished. Cancelling potential speculative or zombie tasks for
this job
cluster-master    | 26/04/05 18:55:43 INFO TaskSchedulerImpl: Killing all running tasks in stage 1: Stage finished
cluster-master    | 26/04/05 18:55:43 INFO DAGScheduler: Job 1 finished: collect at /app/store_index.py:36, took 0.049865 s
cluster-master    | 26/04/05 18:55:43 INFO SparkContext: SparkContext is stopping with exitCode 0.
cluster-master    | 26/04/05 18:55:43 INFO SparkUI: Stopped Spark web UI at http://cluster-master:4040
cluster-master    | 26/04/05 18:55:43 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
```

```
cluster-master    | 26/04/05 18:55:43 INFO TaskSetManager: Finished task 0.0 in stage 1.0 (TID 1) in 30 ms on cluster-master (exe
cutor driver) (1/1)
cluster-master    | 26/04/05 18:55:43 INFO TaskSchedulerImpl: Removed TaskSet 1.0, whose tasks have all completed, from pool
cluster-master    | 26/04/05 18:55:43 INFO DAGScheduler: ResultStage 1 (collect at /app/store_index.py:36) finished in 0.043 s
cluster-master    | 26/04/05 18:55:43 INFO DAGScheduler: Job 1 is finished. Cancelling potential speculative or zombie tasks for
this job
cluster-master    | 26/04/05 18:55:43 INFO TaskSchedulerImpl: Killing all running tasks in stage 1: Stage finished
cluster-master    | 26/04/05 18:55:43 INFO DAGScheduler: Job 1 finished: collect at /app/store_index.py:36, took 0.049865 s
cluster-master    | 26/04/05 18:55:43 INFO SparkContext: SparkContext is stopping with exitCode 0.
cluster-master    | 26/04/05 18:55:43 INFO SparkUI: Stopped Spark web UI at http://cluster-master:4040
cluster-master    | 26/04/05 18:55:43 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
cluster-master    | 26/04/05 18:55:43 INFO MemoryStore: MemoryStore cleared
cluster-master    | 26/04/05 18:55:43 INFO BlockManager: BlockManager stopped
cluster-master    | 26/04/05 18:55:44 INFO BlockManagerMaster: BlockManagerMaster stopped
cluster-master    | 26/04/05 18:55:44 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopp
ed!
cluster-master    | 26/04/05 18:55:44 INFO SparkContext: Successfully stopped SparkContext
cluster-master    | Waiting for Cassandra... attempt 1/30 ({})
cluster-master    | Waiting for Cassandra... attempt 2/30 ({})
cluster-master    | Waiting for Cassandra... attempt 3/30 ({})
cluster-master    | Waiting for Cassandra... attempt 4/30 ({})
cluster-master    | Waiting for Cassandra... attempt 5/30 ({})
cluster-master    | Waiting for Cassandra... attempt 6/30 ({})
cluster-master    | Waiting for Cassandra... attempt 7/30 ({})
cluster-master    | Waiting for Cassandra... attempt 8/30 ({})
cluster-master    | Waiting for Cassandra... attempt 9/30 ({})
cluster-master    | Waiting for Cassandra... attempt 10/30 ({})
cluster-master    | Waiting for Cassandra... attempt 11/30 ({})
cluster-master    | Waiting for Cassandra... attempt 12/30 ({})
cluster-master    | Waiting for Cassandra... attempt 13/30 ({})
cluster-master    | Waiting for Cassandra... attempt 14/30 ({})
cluster-master    | Waiting for Cassandra... attempt 15/30 ({})
cluster-master    | Waiting for Cassandra... attempt 16/30 ({})
cluster-master    | Waiting for Cassandra... attempt 17/30 ({})
cluster-master    | Waiting for Cassandra... attempt 18/30 ({})
cluster-master    | Waiting for Cassandra... attempt 19/30 ({})
cluster-master    | Waiting for Cassandra... attempt 20/30 ({})
cluster-master    | Waiting for Cassandra... attempt 21/30 ({})
cluster-master    | Waiting for Cassandra... attempt 22/30 ({})
cluster-master    | Waiting for Cassandra... attempt 23/30 ({})
```

```
cluster-master    | 26/04/05 18:59:37 INFO ShutdownHookManager: Deleting directory /tmp/spark-5e6925be-787f-48e9-8c37-e0c1b2043dc
4
cluster-master    | 26/04/05 18:59:37 INFO ShutdownHookManager: Deleting directory /tmp/spark-6d2daab7-6123-41e9-803d-a90e4777572
d
cluster-master    | 26/04/05 18:59:37 INFO ShutdownHookManager: Deleting directory /tmp/spark-6d2daab7-6123-41e9-803d-a90e4777572
d/pyspark-7dbbca15-4f01-4172-b32a-0468182e97c0
cluster-master    | Done storing index in Cassandra!
cluster-master    | ==========================================
cluster-master    | Indexing complete!
cluster-master    | ==========================================
cluster-master    | Searching for: This is a query!
cluster-master    | 26/04/05 18:59:39 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using buil
tin-java classes where applicable
cluster-master    | Query: 'This is a query!'
cluster-master    | Terms: ['a', 'this', 'query', 'is']
cluster-master    | Waiting for Cassandra session... 1/36 ({})
cluster-master    | Waiting for Cassandra session... 2/36 ({})
cluster-master    | Waiting for Cassandra session... 3/36 ({})
cluster-master    | Waiting for Cassandra session... 4/36 ({})
cluster-master    | Waiting for Cassandra session... 5/36 ({})
cluster-master    | Waiting for Cassandra session... 6/36 ({})
cluster-master    | Waiting for Cassandra session... 7/36 ({})
cluster-master    | Waiting for Cassandra session... 8/36 ({})
cluster-master    | Waiting for Cassandra session... 9/36 ({})
cluster-master    | Waiting for Cassandra session... 10/36 ({})
cluster-master    | Waiting for Cassandra session... 11/36 ({})
cluster-master    | Waiting for Cassandra session... 12/36 ({})
cluster-master    | Waiting for Cassandra session... 13/36 ({})
cluster-master    | Waiting for Cassandra session... 14/36 ({})
cluster-master    | Waiting for Cassandra session... 15/36 ({})
cluster-master    | Waiting for Cassandra session... 16/36 ({})
cluster-master    | Waiting for Cassandra session... 17/36 ({})
cluster-master    | Waiting for Cassandra session... 18/36 ({})
cluster-master    | Waiting for Cassandra session... 19/36 ({})
cluster-master    | Waiting for Cassandra session... 20/36 ({})
cluster-master    | Waiting for Cassandra session... 21/36 ({})
cluster-master    | Waiting for Cassandra session... 22/36 ({})
cluster-master    | Waiting for Cassandra session... 23/36 ({})
cluster-master    | Waiting for Cassandra session... 24/36 ({})
cluster-master    | Waiting for Cassandra session... 25/36 ({})
```

```
cluster-master    | Waiting for Cassandra session... 26/36 ({})
cluster-master    | Waiting for Cassandra session... 27/36 ({})
cluster-master    | Waiting for Cassandra session... 28/36 ({})
cluster-master    | Waiting for Cassandra session... 29/36 ({})
cluster-master    | Waiting for Cassandra session... 30/36 ({})
cluster-master    | Waiting for Cassandra session... 31/36 ({})
cluster-master    | Waiting for Cassandra session... 32/36 ({})
cluster-master    | Waiting for Cassandra session... 33/36 ({})
cluster-master    | Waiting for Cassandra session... 34/36 ({})
cluster-master    | Waiting for Cassandra session... 35/36 ({})
cluster-master    | Waiting for Cassandra session... 36/36 ({})
cluster-master    | Traceback (most recent call last):
cluster-master    |   File "/app/query.py", line 132, in <module>
cluster-master    |     main()
cluster-master    |   File "/app/query.py", line 48, in main
cluster-master    |     cluster_conn, session = connect_search_session(["cassandra-server"])
cluster-master    |   File "/app/query.py", line 31, in connect_search_session
cluster-master    |     raise RuntimeError("Could not connect to Cassandra: {}".format(last_err))
cluster-master    | RuntimeError: Could not connect to Cassandra: {}
cluster-master    | 26/04/05 19:04:18 INFO ShutdownHookManager: Shutdown hook called
cluster-master    | 26/04/05 19:04:18 INFO ShutdownHookManager: Deleting directory /tmp/spark-aec0b319-d090-4617-9d91-580fd4614c9
8
cluster-master    | Searching for: Automation is fun
cluster-master    | 26/04/05 19:04:20 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using buil
tin-java classes where applicable
cluster-master    | Query: 'Automation is fun'
cluster-master    | Terms: ['automation', 'fun', 'is']
cluster-master    | Waiting for Cassandra session... 1/36 ({})
cluster-master    | Waiting for Cassandra session... 2/36 ({})
cluster-master    | Waiting for Cassandra session... 3/36 ({})
cluster-master    | Waiting for Cassandra session... 4/36 ({})
cluster-master    | Waiting for Cassandra session... 5/36 ({})
cluster-master    | Waiting for Cassandra session... 6/36 ({})
cluster-master    | Waiting for Cassandra session... 7/36 ({})
cluster-master    | Waiting for Cassandra session... 8/36 ({})
cluster-master    | Waiting for Cassandra session... 9/36 ({})
cluster-master    | Waiting for Cassandra session... 10/36 ({})
cluster-master    | Waiting for Cassandra session... 11/36 ({})
cluster-master    | Waiting for Cassandra session... 12/36 ({})
cluster-master    | Waiting for Cassandra session... 13/36 ({})
```